MIXTEC AUSTRALIA
FLUID MIXING TECHNOLOGY

MIXTEC AUSTRALIA
ABN 61 081 675 649
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Malaga, WA, 6090
P.O Box 2599
Malaga, 6944
PH: (08) 9249 2944
FAX: (08) 9249 2955
Website: www.mixtec.com.au
Email: info@mixtec.com.au

DATE : May 12, 2009

NAME : Ben Harrison **MIXTEC REF : A4903**

: Como Engineers **CLIENT'S REF :**

TEL NO : 9432 0100

RE : Randall's Gold

We thank you for your enquiry and have pleasure in submitting our tender for your consideration

The mixers offered have been designed to meet the specified duties in a reliable and efficient manner. While ensuring that process requirements are satisfied, we also pay attention to the mechanical design thus ensuring that the mixer operates without undue noise and vibration and well within it's limits of combined torsional and bending stress.

Please refer to the attached drawings and the attached mixer data sheet for details of our offer.

TERMS AND CONDITIONS

Terms of Payment:
15% on submission of GA drawings
30% on order of gearbox and motors
30% on delivery of gearbox and motors to Mixtec
25% on completion ex works
5% retention of contract sum supplied by Mixtec on dispatch

Price Variation
The prices quoted are fixed and firm for the validity period.

Validity
Our quotation holds good for an cceptance period of 30 days from the date hereof. For period of 90 days Mixtec reserves the right to confirm current pricing

Delivery
The prices given include delivery to Perth Area . We anticipate a delivery period of 26 weeks from receipt of your official order and approval of drawings

Maintenance Instructions and Recommended Spares

Our price includes for 2 sets of our standard maintenance instructions and recommended spare parts lists, together with general arrangement drawings giving full technical details of the equipment offered.

MIXTEC AUSTRALIA

Claims:

The contract is for equipment supply only. There is no Professional Indemnity Insurance. Claims for shortage in delivery or damage in transit cannot be entertained unless we are immediately advised on receipt of goods. Where rubber lining is concerned Mixtec cannot be held liable for lining damage during transport.

Materials of Construction

Mixtec agitators are manufactured from materials which in our opinion, are most suitable for the purpose intended. Since Mixtec has no control over the conditions of use, there is an obligation on the part of the user to check that the material selection is suitable for the installation specified and to use all necessary care and precautions in operation.

Guarantee

The Mixtec heavy-duty mixer gearboxes are fully mechanically guaranteed for a period of one year from the date of commissioning or 18 months from date of delivery, whichever occurs first. This guarantee covers faulty mechanical design, material and/or workmanship. The guarantee work must be carried out in a Mixtec authorised workshop.

Process Guarantee

The units recommended, which are based on data given in your enquiry, will meet or surpass the requirments detailed. Should it fail to do so, at our option, we will modify or replace the unit and/or take it back at full credit.

Under the terms of the guarantee, we are not responsible for any indirect or consequential damages whatsoever, or liability being expressly limited to the equipment we have supplied.

Confidentiality

The purchaser shall not use or communicate to others information derived in writing or orally from us except for the operation, maintenance or improvements of the goods and shall not use the same for the duplication of the goods, nor for a period of five (5) years from the date of acceptance of our quotation, for any extension to the goods, nor the duplication of any part of the goods except with our consent in writing.

Obligations under this clause shall continue after the termination of the contract.

We trust the above meets with your approval, however, should you require any further information, please do not hesitate to contact us.

Yours Faithfully

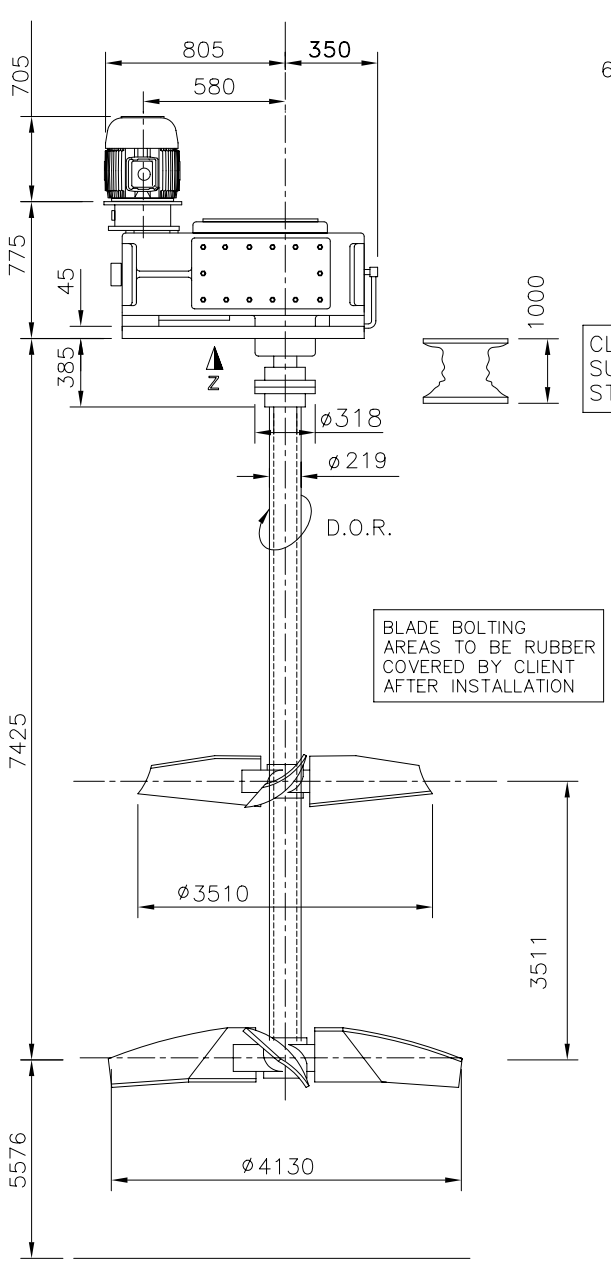
B. BAGULEY
MIXTEC AUSTRALIA

MIXER DATA SHEET			
		MIXTEC AUSTRALIA	
DATE	19 May 09		
NAME	Ben Harrison		
CLIENT	COMO		
PROJECT	Randall's Gold		
TEL NO.	9432 0100		
FAX NO.			
Mixtec REF.	A4889		
CLIENT'S REF.			
TAG NO.	340 AG 01	360 AG 02	
DESCRIPTION.	Leach Tank Agitator No 1	Lime Mixing Agitator	
MODEL No	4009	1077	
DRAWING NO.	A4889/01	A4889/02	
NO. OFF.	1	1	
TANK DIA. (m)	11.8	3	
TANK HEIGHT (m)	12	2.44	
CONTENTS	Slurry	Slurry	
S.G.	1.46	1.12	
VISCOSITY (mPa.s)	10	5	
MATERIAL OF CONSTRN.	C.S.R.C	C.S	
MOTOR (kW)	45	1.5	
MIXER SPEED (r.p.m.)	21	101	
SHAFT DIA (mm)	219	44.5	
SHAFT LENGTH (m)	7425	1.854	
PUMPING RATE (m^3/min)	1477.00	49.55	
INTANK VELOCITY (m/min)	13.50	7.01	
TANK TURNOVERS (per min)	1.20	3.54	
MIXER MASS	3537.00	116.73	
PRICE EACH	$64,937.00	$2,832.00	$0.00
TOTAL PRICE	$64,937.00	$2,832.00	$0.00

Please note the above prices are exclusive of G.S.T.

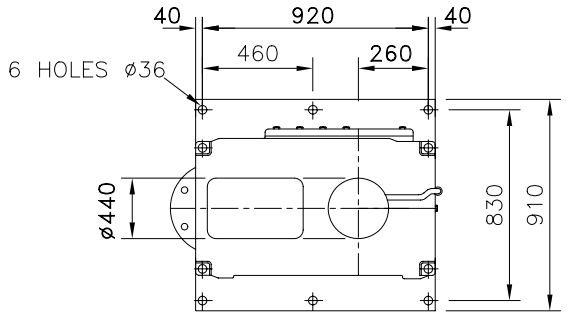
DELIVERY TO SITE	$0.00
TOTAL TENDER PRICE	$67,769.00

CONFIDENTIAL

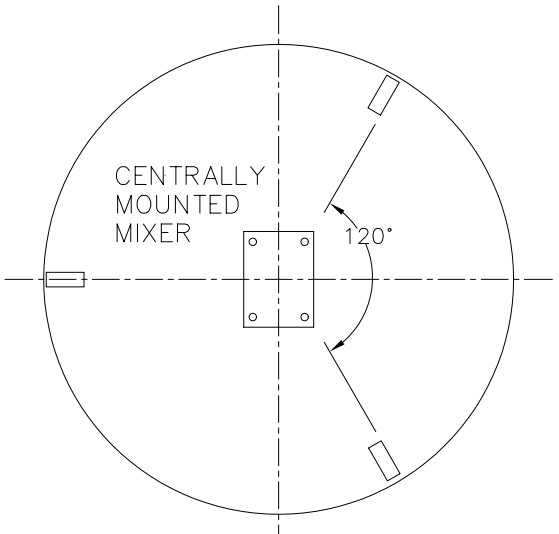


CLIENTS SUPPORT STRUCTURE

BLADE BOLTING AREAS TO BE RUBBER COVERED BY CLIENT AFTER INSTALLATION



VIEW ON ARROW - Z



IN-TANK BAFFLE ARRANGEMENT

MIXER SPECIFICATION

MODEL TYPE	4000
MODEL No.	4009
POSITION	VERTICAL
kW RATING	75
OUTPUT SPEED (RPM)	21
SERVICE FACTOR	1.67
No. OF UNITS	1

MOTOR DETAILS

MOTOR kW	45
MOTOR VOLTAGE	415
MOTOR ENCLOSURE	IP55
FRAME SIZE	225
FRAME MATERIAL	CAST IRON
SPEED (RPM)	1475
SUPPLIED BY	MIXTEC

OTHER INFORMATION

MIXER
MOTOR
PAINTING
PACKING
TESTING

SERIAL NUMBER

TANK INFORMATION

DIAMETER (m)	11.8
HEIGHT (m)	12
TOP	OPEN
BOTTOM	FLAT
BAFFLE WIDTH	983
MIN TANK OPENING	4130
SEALING	NO
TYPE	NONE
PRESSURE (kPaG)	0
TEMPERATURE (°C)	45

TANK DESIGN LOADS

MIXER MASS (kg)	3475
BEND. MOMENT(Nm)	65129
TORQUE (Nm)	61393
AXIAL LOAD (N)	51135

TANK CONTENTS

PRODUCT	SLURRY
S.G.	1.46
VISCOSITY (mPa.s)	10

CONSTRUCTION MAT'LS

COUPLING	C.S.
SHAFT	C.S
IMPELLERS	C.S
SHAFT COATING	
IMPELLER COATING	S.N.R

IMPELLER DETAILS

	LOWER	UPPER
IMPELLER TYPE	HA715	HA800
BLADES	BOLTED	BOLTED
HUB	WELDED	WELDED
STABILISERS	NO	NO

CLIENT DETAILS

ORDER NUMBER

TAG NUMBER 340 AG 01

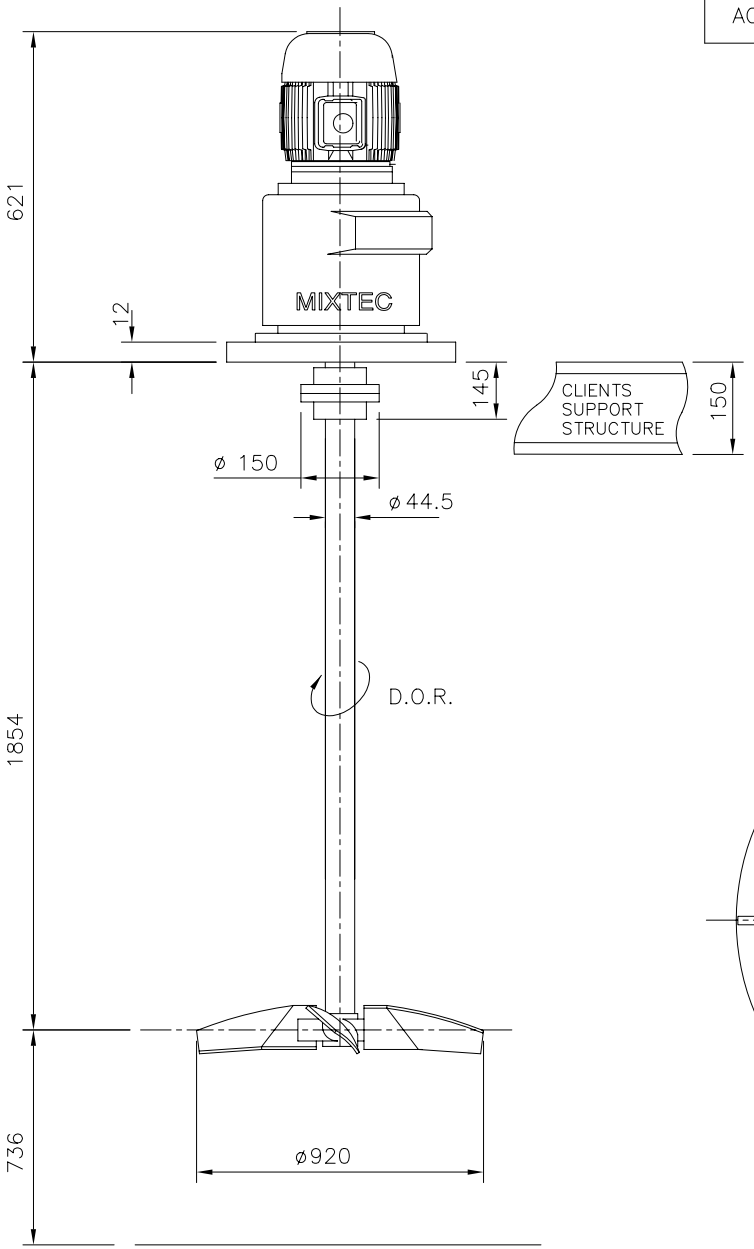
TECHNOLOGY TOP ◆ 100
SOUTH AFRICAN TECHNOLOGY FOR WORLD MARKETS 1991

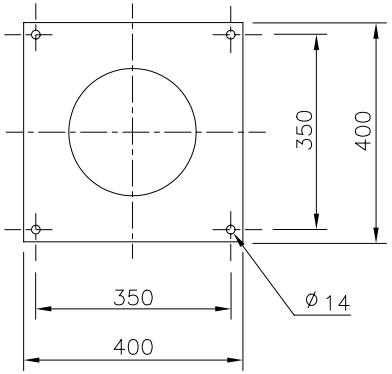
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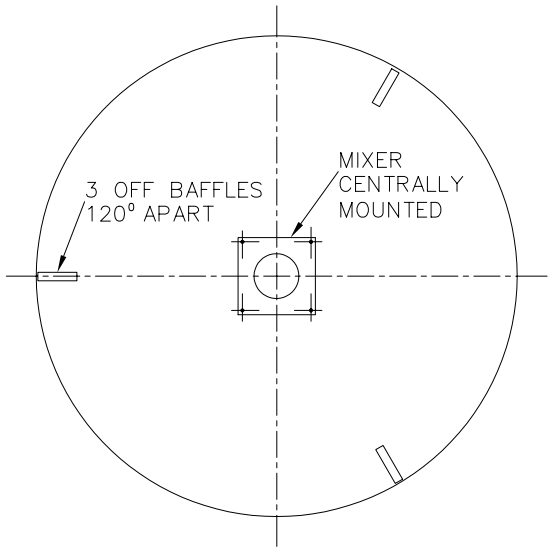
A		B		NOT TO SCALE	CLIENT APPROVAL
DRAWN	B. BAGULEY	APPROVED		CLIENT: COMO	
CHECKED		POSITION		PROJECT	
DATE	12 May 2009	DATE		LEACH TANK No 1	
				DRG. No.	A4903/01

ADAPTER WITH FLEXIBLE COUPLING TO ACCEPT STANDARD FLANGED MOTOR FITTED





BASEPLATE DETAIL



IN-TANK BAFFLE ARRANGEMENT

MIXER SPECIFICATION

MODEL TYPE	1000
MODEL No.	1077
POSITION	VERTICAL
kW RATING	7.67
OUTPUT SPEED (RPM)	101
SERVICE FACTOR	5.11
No. OF UNITS	1

MOTOR DETAILS

MOTOR kW	1.5
MOTOR VOLTAGE	415
MOTOR ENCLOSURE	IP55
FRAME SIZE	90
FRAME MATERIAL	CAST IRON
SPEED	1420
SUPPLIED BY	MIXTEC

OTHER INFORMATION

MIXER	
MOTOR	
PAINTING	
PACKING	
TESTING	SHOP RUN

MIXTEC DETAILS
SERIAL NUMBER

TANK INFORMATION

DIAMETER (m)	3
SHELL HEIGHT (m)	2.44
TOP	OPEN
BOTTOM	FLAT
BAFFLE WIDTH	250
MIN TANK OPENING	920
SEALING	NO
TYPE	NONE
PRESSURE (kPaG)	0
TEMPERATURE(°C)	20

TANK DESIGN LOADS

MIXER MASS (kg)	117
BEND. MOMENT(Nm)	716
TORQUE (Nm)	425
AXIAL LOAD (N)	1718

TANK CONTENTS

PRODUCT	Slurry
S.G.	1.12
VISCOSITY (mPa.s)	1

CONSTRUCTION MAT'LS

BASE PLATE	C.S.
COUPLING	C.S.
SHAFT	C.S
IMPELLERS	C.S

IMPELLER DETAILS

IMPELLER TYPE	HA715
BLADES	BOLTED
HUB	WELDED
STABILISERS	NO

CLIENT DETAILS

ORDER NUMBER

TAG NUMBER 360 AG 02

TECHNOLOGY TOP ◆ 100
SOUTH AFRICAN TECHNOLOGY FOR WORLD MARKETS 1991

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A		B		NOT TO SCALE	CLIENT APPROVAL
				DRAWN: B. BAGULEY	APPROVED:
				CHECKED:	POSITION:
				DATE: 8 May 2009	DATE:

CLIENT: COMO

TITLE: Lime Mixing Agitator

DRG. No.: A4903/02

MIXTEC

FLUID MIXING TECHNOLOGY

2020



PERFECT VISION INTO GOOD MIXING PRACTICE

75 Years of experience and know how



132kW Series 4015 Mixer Drive

Profile

Mixtec International has seven manufacturing plants strategically placed around the world to give you, our customer, the service and fluid mixing technology that you require, when you require it.

Each plant has its own research and development as well as design, engineering and commissioning personnel who are trained to solve your mixing problems with the latest fluid mixing technology.

For 75 years, the Mixtec group has been engaged in mixing technology. Since 1930 as Kestner Industries, UK Mixtec has been committed to quality, reliability and service



Mixtec Agitator shaft coupling and support collar



4000m³
17.0m Dia Gold Leach tanks



Dual 6.0m Dia HA715 Impellers

Whether your requirements are simple blending of solids suspension to the more exacting standards of multiple complex reactions, Mixtec has the capability and experience to suit your needs.

Mixtec's proven track record of thousands of installations around the world bears witness to our success and our commitment to good mixing practice.

Good Mixing practice starts with product research and development



This enables Mixtec to produce custom made equipment and cost effective solutions to your mixing problems.

Our laboratory can guide you in the correct selection of impeller systems as well as in-tank items such as Sparge ring design and baffle arrangements.



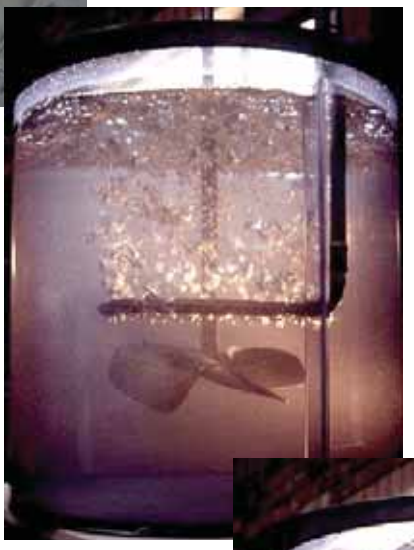
Complex fluid flow and gas dispersion test rig.

Research into optimum tank characteristics-baffling-inlet-outlet configuration and optimisation of gas sparge.

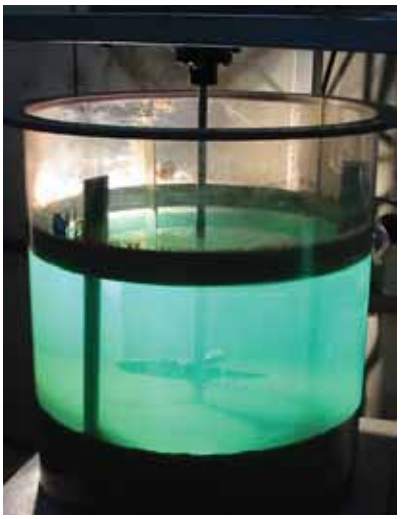
Over a period of many years Mixtec has developed new impeller systems and improved on existing agitator duties in all major industries.



Gold Leach test work

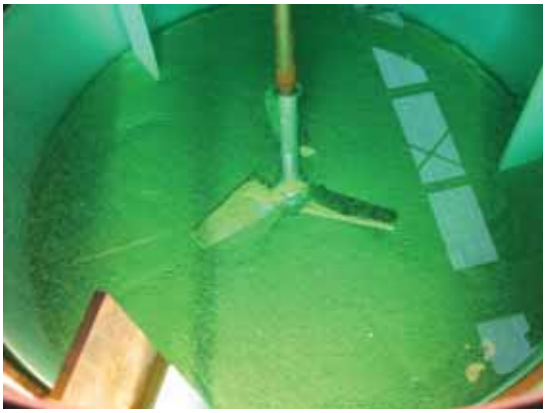


New gas dispersion impellers and sparge rings under test



Solid suspension test work on Copper leaching application.

Using established scale up procedures accurate and predictable process results are obtained.



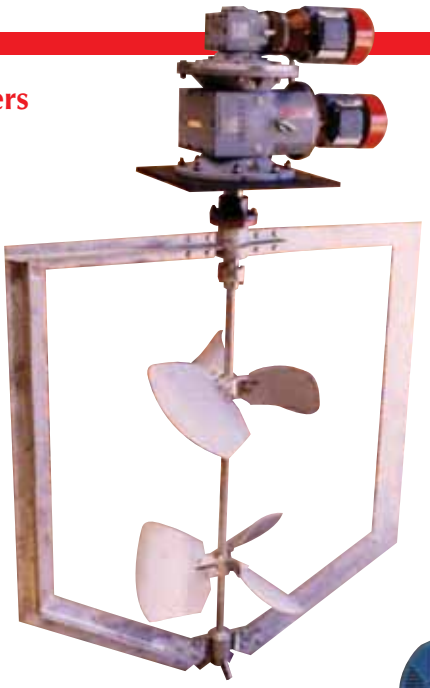


Mixtec is in constant contact with operators and all important research institutes around the world in order to offer the very best solution to your agitation problems

Custom designed mixers



Magnetic drive for Bio processing



Contra rotating mixer impellers for viscous blending duty

All mixing applications are different and each has its own unique solution.

At Mixtec we understand this and our engineers strive to improve every aspect of mixer selection.



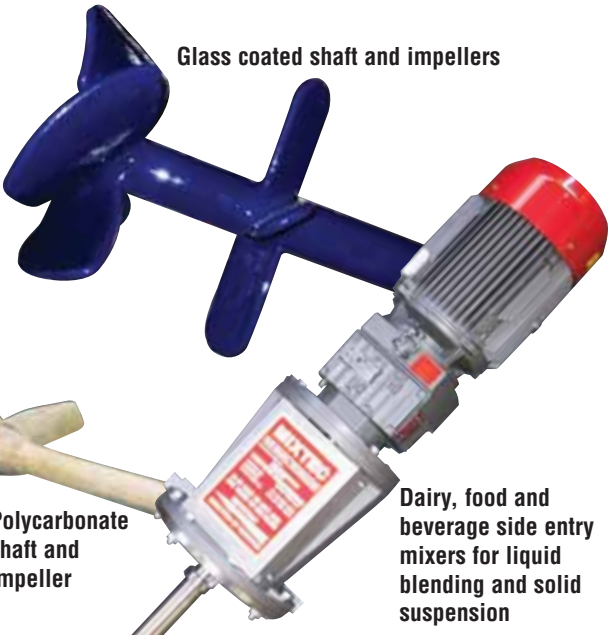
Series 5500 Improved side entry agitators for paper and pulp applications

Mixtec's main activities include:

- The research and development of mixing technology
- Processed design and construction of fluid mixers
- Comprehensive mechanical design of fluid mixing components
- Design and development of alternative agitation systems
- Design and development of fluid pumping devices.



Special bottom entry agitator for crystalliser duty



Glass coated shaft and impellers

Polycarbonate shaft and impeller

Dairy, food and beverage side entry mixers for liquid blending and solid suspension



600mm Dia static mixers

Good Mixing practice - selecting the right drive for the job



Mixtec Series 2000 drives





Series 5000 side entry agitator for petro-chemical application

Features include:

- Oversized bearings to take start-up shock
- Dry well construction - no oil leaks
- Large output shafts, robust construction
- Hardened and ground gears to AGMA 2000 standards for maximum strength and wear and reliability



Robust drive casings on the Series 4000 after stress relieving and NDT testing



Mixtec Series 3000 right angle drive under test

Mining and Refining Agitators for all duties



Cobalt, Tin, Aluminium, Lead, Mineral Sands and others



Gold, Platinum. Uranium, Silver, Copper, Nickel, Zinc





In 1980 thirty 75kW draft tube agitators were supplied to Rand Mines Milling and Mining for the processing of Gold Dumps around Johannesburg, South Africa.

Using the new CIP Adsorption technology these units have now treated over 50 million tonnes of rock resulting in the recovery of over 40,000kg of Gold.







Mixtec has extensive knowledge of solvent extraction pump secondary and auxillary mixer duties.



75kW Precipitation Reactor

Process duties require an efficient and reliable solution. Mixtec has the track record, technology and equipment.

Chemical and Industrial mixing applications



Chemical processing has some of the most varied and difficult agitation problems.



Series 4000 agitators on a Resin plant installation. UK



Chemical plant. Malaysia



Series 1000 units on paint blending duty



Special solvent extraction column with 28 stage counter current agitation



Series 2000 agitator on a crystalliser duty



Large HA 735 impeller in a high pressure, high temperature autoclave

Mixtec has the experience and know-how to solve your mixing needs

Dairy Food, Beverage and Pharmaceutical industry





Series 2000 on a soup cooker.



Food blending application



One of 104 Series 5000 side entry agitators supplied to a large winery. New Zealand

High quality finish to latest hygiene standards



Multi stage fermenter hygienic designed surfaces for CIP processes



Also available:

- In-line static mixers
- Jet and eduction mixing
- High viscosity blending
- Sugar pan circulators
- High Speed dispersion





Series 1000 on flocculator duty - USA







1992 major installation of anoxic and anaerobic agitators.

For over 30 years Mixtec has supplied thousands of agitators to the Water Treatment industry

Side mounted Series 1000 - Australia

Anoxic Mixer

Clean water is precious and water care is rapidly becoming one of the worlds major problems.

Mixtec has been at the forefront of many water treatment processes due to its continual research and development.

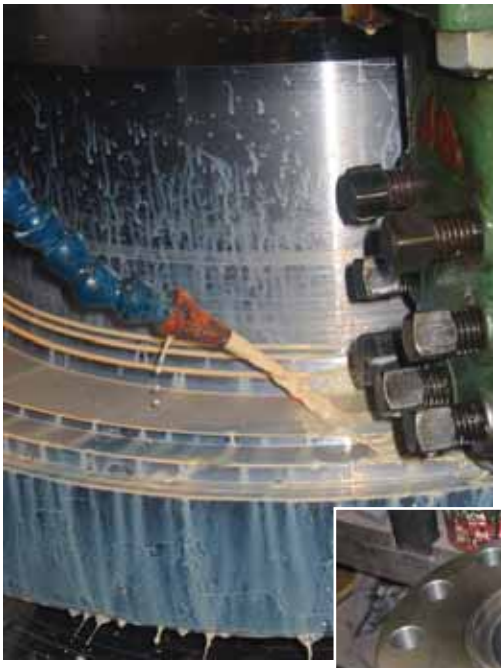


Series 2000 on aeration basin - South Africa



Series 1000 with weather protection and VVVF control

Engineering and Fabrication





Large agitator couplings being manufactured to tolerances of less than 0.01mm for accuracy and control.



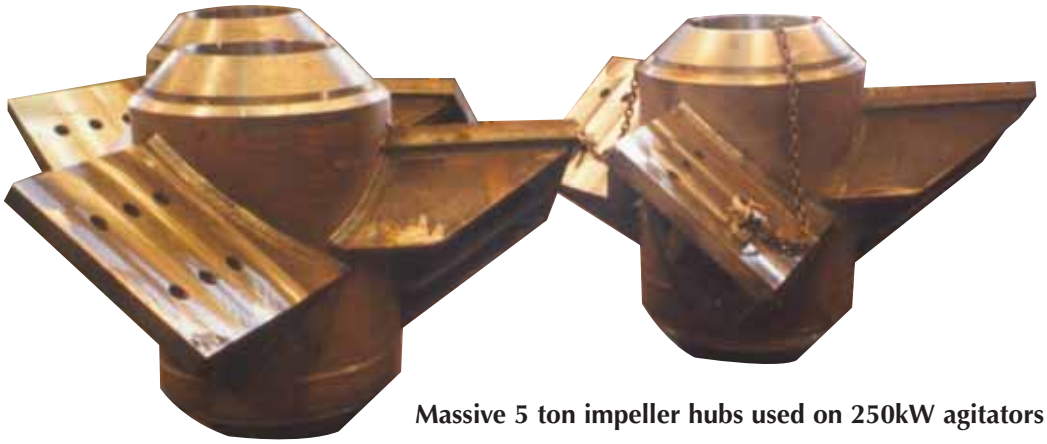
All agitators are manufactured using the latest international standards for mixer design, to produce a rugged well constructed agitator backed by a solid mechanical and process guarantee.





Series 5500 side entry

Complete assembly of all components before despatch



Massive 5 ton impeller hubs used on 250kW agitators



500mm nominal bore schedule 80 pipe shaft in production. Shaft length exceeds 20 metres

Mixtec has a wealth of experience which covers all aspects of agitator construction.

Our computer designs not only check process parameters but also calculate bending and torsional loads.

This information is used to design material sizes and fabrication techniques.



NDT inspection on reinforced impeller hub before magnetic particle testing and final coating



All factors influencing the mechanical design are analysed and backed up by years of successful experience in the field.

Materials of construction include Carbon and Stainless steels, Hastelloy, Incalloy, Tantalum and Titanium plus several varieties of coatings such as rubber, stellite, Halar, glass and poly-carbonates.



Final test on Series 4000 in 5m Dia test tank

Around the world with Mixtec - We are committed to you



Chemical plant - Brazil



Gold plant - South Africa



Zinc storage - Namibia



Back fill - Indonesia



Beverage storage - Australia



Mineral Leach tank - USA



Water treatment - China



Paint plant - New Zealand



Paper and pulp side entry mixers - UK

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